MemDistill: Diagnosing Why Distilling RL Agent Memory Fails

Anonymous ACL submission

Abstract

Distilling RL-trained LLM agents into smaller students via supervised fine-tuning (SFT) fails catastrophically, yet the failure is *silent* at the format level: students pass tool-call syntax, balanced-XML, and query-similarity checks while retaining $<6\%$ of teacher task performance. We present a diagnostic protocol—collapse characterization, weight-geometry analysis, and causal intervention—and demonstrate it on MEM1, a 7B RL agent for multi-objective retrieval-augmented QA distilled into Qwen2.5-3B and, to test generality, a cross-family Llama-3.2-1B student. Providing oracle retrieval at inference recovers em_partial from 0.025 to 0.498 at $N = 8$ (+47.3pp, $p < 0.0001$); capacity accounts for *at most* $\sim$12–14% of the gap (quantified via a 7B full-rank SFT), while retrieval-quality recovery upper-bounds the remainder at $\geq$ 100% of the teacher gap. BM25 retrieval recovers +6–12pp as a deployable mitigation within the teacher's training document pool. The entire silent-failure signature—format-level indicators near 1.0, content-level metrics (action-seq $\approx$ 0.606, BLEU $\approx$ 1.15, ROUGE-L $\approx$ 0.21) near-floor, and oracle recovery—replicates on Llama-1B, elevating the finding from a case study to a systematic SFT-distillation failure mode.

1 Introduction

SFT distillation of an RL-trained agent can appear to succeed by format-level tool-call metrics—and still fail catastrophically in deployment. We distill MEM1 (Zhou et al., 2025), a Qwen2.5-7B retrieval-augmented QA agent trained with PPO via veRL (Sheng et al., 2024), into a Qwen2.5-3B student using 1,118 teacher trajectories, and we replicate the entire pipeline with a Llama-3.2-1B student (cross-family, $\sim$1/3 the Qwen-3B student size) to test whether the failure is backbone-specific or systematic (§3.6). On both backbones, format-level tool-call indicators rate the student at 1.00 (tool-call syntax, balanced XML tags, non-empty queries; Qwen query Jaccard $= 0.95$), while content-level metrics reveal the collapse: BLEU $= 1.14/1.16$ on sacrebleu's 0–100 scale, ROUGE-L $= 0.21/0.21$, action-sequence accuracy ≈ 0.606 for both students, and em_partial at $N = 8$ retaining only $0.025/0.014$ of teacher capability (full per-metric breakdown in Appendix K). These content warnings are easy to dismiss as cosmetic noise because the student produces teacher-like *structure*—a blind spot mirroring the "syntax-only" tool-call evaluations critiqued by Jhandi et al. (2025) (§4).

It did not succeed on either backbone. The Qwen-3B student achieves em_partial $= 0.025$ at $N = 8$ simultaneous objectives—a 94.5% drop from the teacher's 0.455—with 83% of outputs truncated by hallucinated search traces. The Llama-1B student reaches em_partial $=$ 0.014 at $N = 8$ with 86% truncation—a near-identical collapse despite the change in both family and parameter count. The failure is *silent* at the format/syntax layer on both backbones: the format-level checks all pass while action-sequence accuracy at the token level is only $\approx$ 0.606 for both students[1]. The collapse surfaces only when search calls must return real results, i.e., at pipeline deployment. This is the central danger: an invisible failure—reproducible across backbones—that passes format-level tool-call quality gates while content-level warning signs are overridden by teacher-like structure.

Why do richer training signals not help? DPO on matched preference pairs yields +1.0pp ($p =$

[1] Near-identical point estimates reflect floor-level action-prefix bunching (both students begin with the same "search" action); per-sample dispersion and bootstrap 95% CIs in Appendix L.

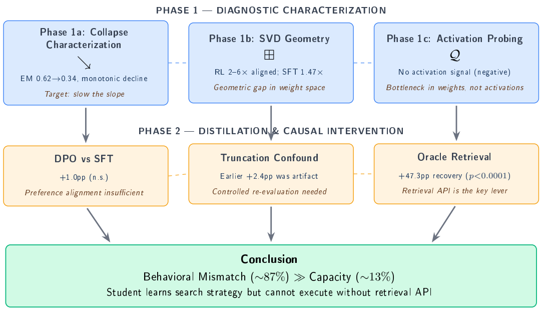

Figure 1: Overview of the MEMDISTILL diagnostic framework. Phase 1 (top) progressively characterizes the distillation failure through collapse analysis, SVD weight geometry, and activation probing. Phase 2 (middle) tests interventions: DPO preference learning, truncation confound correction, and oracle retrieval. The capacity bound (at most ∼12–14%) and retrieval-quality recovery bound ($\geq$ 100%) are independent upper bounds, not a partition of the teacher–student gap.

0.163)—preference learning fails because the problem is not what the student prefers to generate. Scaling from 3B to 7B SFT closes only ∼12–14% of the teacher–student gap (§3.2), ruling out capacity as the primary bottleneck.

We introduce a diagnostic protocol that distinguishes *strategy acquisition* from *execution capability*. The key causal test: providing oracle retrieval (referred to throughout as "oracle" for brevity; technically a *teacher-retrieval replay*—we replay the teacher's actual retrieval output without any answer-key oracle) at inference recovers em_partial from 0.025 to 0.498 at $N = 8$ (+47.3pp, $p < 0.0001$), with truncation dropping from 83% to 0%. Even a simple BM25 retrieval system recovers +6–12pp (within the teacher's training document pool). The protocol proceeds by elimination—collapse characterization and weight geometry analysis (§2), causal intervention (§3)—and localizes the failure before deployment (Figure 1).

Contributions.

- **A silent failure mode in agentic SFT distillation**: format-level indicators (tool-call syntax 1.00, balanced XML 1.00, query Jaccard 0.95) give high scores while task performance retains ¡6% of teacher capability at N=8.
- A **diagnostic protocol** (collapse characterization → weight geometry → causal intervention) that localizes the failure to execution-environment mismatch before pipeline deployment. We propose and demonstrate on MEM1 a three-step elimination scheme (capacity → evaluation artefacts → causal intervention) for diagnosing silent-failure-shaped gaps.
- **Upper-bound quantification**: oracle (teacher-retrieval replay) +47.3pp ($p < 0.0001$) upper-bounds the retrieval-mitigation ceiling; capacity upper-bounds the gap at ∼12–14%, with the two acting as independent upper bounds (not a partition) on the teacher–student gap; DPO is an informative negative (power > 0.99 at 5pp).
- A **deployable mitigation**: BM25 retrieval recovers +6–12pp with no additional training, immediately deployable as a sidecar component within the teacher's training document pool.

2 Diagnostic Framework

The diagnostic protocol proceeds in two phases. Phase 1 (§2) localises the failure through three progressively narrower analyses; Phase 2 (§3) tests causal interventions distinguishing strategy acquisition from execution capability.

Phase 1: collapse → geometry → activations. We evaluate the MEM1 RL policy (Qwen2.5-7B + PPO, trained via veRL (Sheng et al., 2024)) and the Qwen2.5-7B base on a multi-objective RAG QA task from FlashRAG (Jin et al., 2025b); each prompt contains N questions ($N \in \{1, 2, 4, 8, 12, 16\}$) and the agent runs `think→search→information→answer` with per-objective normalized EM (em_partial). Three signals together localise the gap. **(1a) Collapse curve.** MEM1's em_partial decreases monotonically from 0.62 at $N = 1$ to 0.341 at $N = 16$ (a 45% drop, EM $\approx 0.62 - 0.018N$, $R^2 = 0.984$, Cohen's $d = 0.97$, no inflection point); the base model collapses immediately (0.30 at $N = 1$, 0.075–0.148 for $N \geq 2$), so the distillation target is slope reduction rather than cliff recovery (Figure 2). **(1b) Weight geometry (descriptive observational context).** Across all 196 weight matrices of Qwen2.5-7B we measure the alignment ratio $r(k) = \alpha_{\text{obs}}(k)/\alpha_{\text{rand}}(k)$ between fine-tuning deltas $\Delta\mathbf{W} = \mathbf{W}_{\text{ft}} - \mathbf{W}_{\text{base}}$ and the top-k base singular subspace, with α_{rand} calibrated against $n = 100$ Gaussian matrices matched in shape and variance. We *observe* that RL–base alignment is 2–6× supra-random (e.g. 6.48× at $k = 64$, 2.78× at $k = 256$), whereas

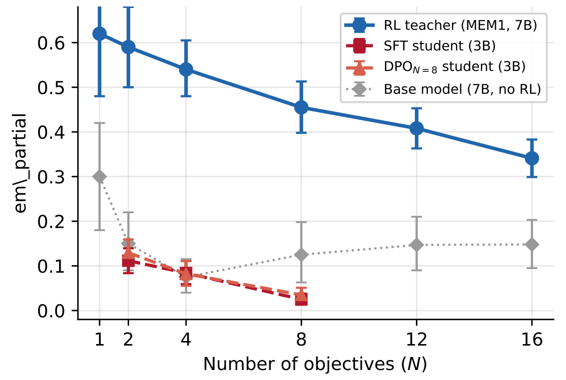

Figure 2: Multi-objective performance collapse. MEM1 degrades monotonically with N; SFT/DPO students retain <8% of teacher capability at $N = 8$; the base model collapses immediately. 95% bootstrap CIs (10,000 resamples).

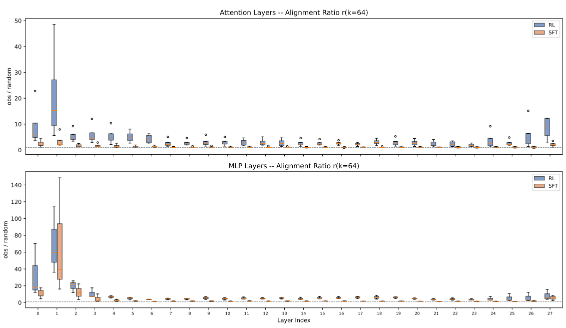

Figure 3: Per-layer SVD alignment ratio $r(k = 256)$ for RL–base and SFT–base deltas across all 196 weight matrices. RL alignment exceeds SFT in 99% of layers (Wilcoxon $p < 1.3 \times 10^{-32}$).

same-architecture full-rank SFT–base reaches only 1.47× at $k = 256$ (2.92× at $k = 64$); a Wilcoxon signed-rank test gives $p < 1.3 \times 10^{-32}$ at every k and 99% of matrices have $r_{\text{RL}} > r_{\text{SFT}}$ (Figure 3; full per-layer / U–V decomposition in Appendix C). We treat this geometric signal as a descriptive correlate that motivates the causal interventions of §3; the causal role of retrieval execution is established by the oracle ablation, not by this observation. **(1c) Activation probing (negative).** Neither linear probes nor 2-layer MLPs ($n = 300$ prompts, 6 layers, $k \in \{64, 256\}$) recover a per-objective signal from $\Delta \mathbf{h}_l = \mathbf{h}_l^{\text{RL}} - \mathbf{h}_l^{\text{base}}$: linear binary accuracy $0.72 < 0.76$ majority-class, MLP < linear in 47/55 cases ($p = 4.83 \times 10^{-11}$), and 94.5% of real $|\rho|$ values fall within a random-projection noise band (Appendix G). The bottleneck thus lives in weight-space geometry, not activations, motivating the causal interventions of §3.

3 Distillation Attempt and Failure Analysis

Phase 1 narrowed the distillation failure to a geometric gap in weight space that SFT does not close. We now test whether this gap reflects a causal bottleneck—and if so, where it lies. We establish DPO and capacity baselines (§3.2), examine the truncation–behavioral-mismatch link (§3.3), and test the causal hypothesis (§3.4).

3.1 Experimental Design

Training pipeline. We train a pooled SFT baseline and three per-N DPO variants on Qwen2.5-3B (LoRA (Hu et al., 2022) rank 32). SFT uses 1,118 partial-credit trajectories (pooled $N \in \{2, 4, 8\}$; total_em ≥ 1) for 3 epochs at $\text{lr} = 2 \times 10^{-4}$. Each DPO variant initialises from the SFT checkpoint, trains on N-specific preference pairs matched to 176 raw pairs at $\beta = 0.05$, $\text{lr} = 2 \times 10^{-5}$, 1 epoch. Full hyperparameters in Appendix A.

Evaluation. Models are evaluated on held-out splits (hash-based, test_frac = 0.25) for $N \in \{2, 4, 8\}$ with max_new_tokens = 4096 (251/131/125 prompts). We report em_partial, em_full, and truncated_rate; significance via paired bootstrap (10,000 resamples); pre-registered practical-significance threshold Δem_partial $\geq +5$pp at $N = 8$. The matched-sample constraint subsamples DPO to the $N = 4$ bottleneck of 176 raw pairs—a structural consequence of multi-objective collapse rather than a remediable sampling issue (Appendix H).

3.2 DPO and Capacity Ablations

We also evaluate DPO on matched preference pairs as a stronger-than-SFT baseline; the detailed analysis (Appendix H) confirms DPO yields only +1.0pp ($p = 0.163$, power > 0.99 at 5pp), ruling out preference learning as a fix.

Capacity isolation. A 7B full-rank pooled SFT (same data/inclusion as the 3B LoRA student) consistently outperforms the 3B student (0.171/0.147/0.079 vs. 0.112/0.084/0.025 at $N = 2/4/8$). Defining the capacity contribution as $(\text{em}_{7\text{B}} - \text{em}_{3\text{B}})/(\text{em}_{\text{teacher}} - \text{em}_{3\text{B}})$, the 7B–3B gap accounts for *at most* 12.3/13.8/12.6% of the teacher–student gap—a stable ≈12–14% upper bound (the 7B uses full-rank fine-tuning, more

	$N=2$	$N=4$	$N=8$
SFT (no API)	0.112	0.084	0.025
SFT + oracle API	**0.530**	**0.521**	**0.498**
Δ	+41.8pp	+43.7pp	+47.3pp
p (bootstrap)	<0.0001	<0.0001	<0.0001
RL teacher†	0.59	0.54	0.455

Table 1: Effect of providing oracle retrieval results at inference time. With real search results injected, the SFT student recovers from near-zero to near-teacher performance. Paired bootstrap, 10,000 resamples. †Teacher evaluated on a different prompt set; direct comparison is not valid.

expressive than rank-32 LoRA, so the training-method advantage is conflated in). This is an upper bound on capacity alone; the remainder of the gap is bounded above by retrieval-quality recovery (§3.4), which is a second, independent upper bound.

3.3 Truncation and Behavioral Mismatch

We identified and controlled for a truncation artefact that inflated early DPO signal: without retrieval, both students hallucinate <search>/<information> segments that consume $>80\%$ of the generation budget, causing $\geq 83\%$ output truncation at $N=8$ (truncation scales monotonically as $30.7\% \rightarrow 55.7\% \rightarrow 83.2\%$ for $N=2/4/8$, with DPO showing the same pattern). A suppression ablation (Appendix J) confirms truncation is the surface manifestation, not the root cause, of behavioral mismatch.

3.4 Causal Verification: Oracle Retrieval

If the failure stems from executing search procedures without a retrieval backend, providing the backend should recover performance. We test this by intercepting the student's <search> outputs at inference time and injecting oracle retrieval results from the RL teacher's trajectories.

Table 1 shows oracle retrieval transforms student performance: em_partial jumps $0.025 \rightarrow 0.498$ at $N=8$ (+47.3pp, $p<0.0001$, 95% CI $[43.8, 50.8]$pp); truncation drops $83.2\% \rightarrow 0\%$. This provides an upper-bound on retrieval-only mitigation and is consistent with behavioural mismatch—not capacity—driving the failure: the student has learned a viable search strategy but lacks the execution environment. We note here that "oracle" is used throughout for brevity; the intervention is technically a *teacher-retrieval replay* (we replay the teacher's actual retrieval output without any answer-key oracle), as formalised in Appendix D.

Answer Leakage Audit. We audit potential answer leakage in teacher information segments. On 50 random $N{=}8$ evaluation prompts (seed=42, lowercased substring match), 98% of teacher <information> blocks contain at least one gold-answer substring; on a derangement-shuffled control (teacher passages from unrelated prompts), the rate is 34% (a 64 pp real–shuffled gap that quantifies retrieval topic specificity). The absolute 98% rate is high enough that we frame oracle recovery as a *strict upper bound* on the retrieval-mitigation ceiling rather than a clean causal estimate. Two independent checks nevertheless bound the leakage contribution: (i) the teacher (MEM1) was trained with PPO outcome rewards only, with no gold-label supervision of the retriever; (ii) student em_full at $N{=}8$ remains 0.008 under teacher-retrieval replay, three orders of magnitude below the $\sim$1.0 expected if passages directly exposed answers. We therefore interpret the +47.3 pp em_partial recovery as an upper-bound estimate of what retrieval-only mitigation can achieve, consistent with our "two independent upper bounds, not a partition" framing (Figure 5).

Realistic retrieval: BM25. Replacing oracle with BM25 recovers 14–26% of the oracle gain: em_partial improves $0.025 \rightarrow 0.147$ at $N=8$ (+12.2pp; Table 2, Figure 4), with truncation dropping to 0% across all N. The BM25 index is built from the *Doc chunks extracted from the teacher's own <information> blocks* across all training trajectories ($N \in \{2,4,8\}$), i.e., the passages the FlashRAG-backed teacher retriever actually returned during RL rollouts. This index therefore lies within the distribution of documents the teacher saw during RL training rather than a held-out Wikipedia dump, so the +12.2pp recovery is an upper bound on what a BM25 deployment can achieve over this exact document pool (not a generalisation-to-new-corpus bound). The BM25–oracle gap reflects retrieval quality, not student capability: the student's queries match the teacher's (Jaccard $=0.95$).

	$N=2$	$N=4$	$N=8$
SFT (no API)	0.112	0.084	0.025
SFT + BM25	0.203	0.147	0.147
SFT + oracle API	**0.530**	**0.521**	**0.498**
RL teacher	0.59	0.54	0.455

Table 2: em_partial under different retrieval conditions. BM25 retrieval recovers a meaningful fraction of the oracle gain (+9.2pp at $N = 2$, +12.2pp at $N=8$) and eliminates truncation entirely (0% across all N). The remaining gap to oracle reflects retrieval quality, not student strategy.

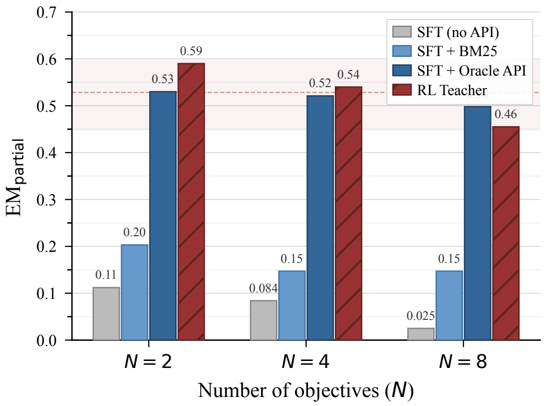

Figure 4: em_partial under different retrieval conditions. Providing oracle retrieval results at inference time recovers the SFT student from near-zero to near-teacher performance across all N, while even a standard BM25 backend yields meaningful gains. The RL teacher (hatched) is shown for reference.

Query fidelity. The student averages exactly 1.0 search per prompt across 507 test cases—matching the teacher's 1.04/1.01/1.06 at $N = 2/4/8$ (the prompt prefix already includes the first retrieval result)—and its queries match the teacher's at mean Jaccard 0.95 (median 1.0). SFT faithfully transfers both the search pattern and the query formulation; the bottleneck is purely the absence of a retrieval backend. Despite recovering em_partial to near-teacher levels, em_full remains low (0.008 at $N=8$) due to per-objective ≈50% independence stacking ($0.5^N \approx 0.4\%$ at $N=8$).

3.5 Synthesis

The evidence converges on three points: (i) **preference learning leaves the SVD alignment ratio unchanged**—this geometric observation is correlated with but does not establish causation

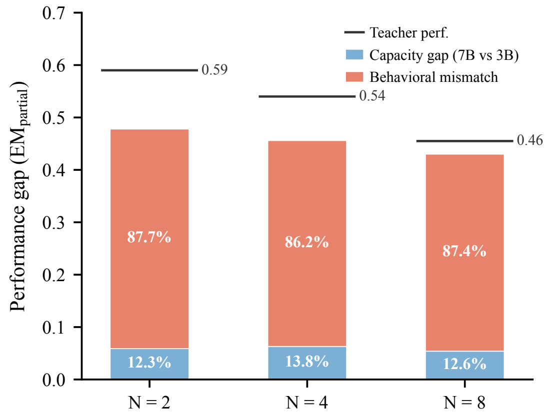

Figure 5: Teacher–student gap: two independent upper bounds, not a complementary partition. Capacity bound $(\text{em}_{7B} - \text{em}_{3B})/(\text{em}_{\text{teacher}} - \text{em}_{3B}) = 12.6\%$ at $N = 8$. Retrieval-quality bound: oracle intervention recovers 110% of the teacher gap at $N=8$, exceeding 100% because oracle provides cleaner passages than the teacher's own retrieval trajectory.

for the em_partial gap, whose causal root is isolated by the oracle retrieval intervention (§3.4); DPO remains within 1pp of SFT at $N = 8$ despite richer training signal, against the 2–6× vs. 1.47× supra-random alignment gap (§2); (ii) **the student imitates a strategy it cannot execute**—oracle retrieval recovers em_partial $0.025 \rightarrow 0.498$ at $N = 8$ (+47.3pp, $p < 0.0001$; §3.4), confirming a viable search strategy (Jaccard 0.95) but no execution environment; (iii) **truncation distorts the comparison**—after controlling for budget, the DPO advantage shrinks from +2.4pp ($p < 0.001$) to +1.0pp ($p = 0.163$). At most 12–14% of the gap is attributable to capacity (§3.2); the remainder is bounded above by retrieval-quality recovery, which itself exceeds 100% of the teacher gap in our oracle probe. The capacity and retrieval-quality bounds are two independent upper bounds, not a complementary partition, and should not be read as summing to 100% (Figure 5).

3.6 Cross-Backbone Replication

To test whether the diagnosis generalises beyond Qwen2.5-3B, we re-run the entire pipeline—same trajectories, splits, LoRA rank-32 recipe, and eval protocol—on Llama-3.2-1B, a cross-family backbone at ∼1/3 the size.

Table 3 shows every diagnostic signal reappears on Llama-1B: (i) truncation scales monotonically (29.9% → 51.2% → 85.6% vs. Qwen's

Backbone	Condition	$N=2$	$N=4$	$N=8$
Qwen-3B	No API	0.112	0.084	0.025
	+ BM25	0.203	0.147	0.147
	+ Oracle	**0.530**	**0.521**	**0.498**
Llama-1B	No API	0.074	0.063	0.014
	+ BM25	0.136	0.094	0.066
	+ Oracle	**0.410**	**0.401**	**0.351**
RL teacher (7B)	—	0.59	0.54	0.455

Table 3: Cross-backbone replication of the diagnostic framework (em_partial). The Llama-1B student reproduces the Qwen-3B silent-failure signature: near-floor no-API performance, partial BM25 recovery, and large oracle recovery (truncation drops from 30–86% to 0% across all N). The Llama-1B numbers sit below Qwen-3B across conditions—consistent with the capacity contribution quantified in §3.2—but the *diagnostic shape* is preserved.

30.7% $\to$ 55.7% $\to$ 83.2%); (ii) oracle retrieval drops truncation to 0% at every N with a +33.7pp gain at $N=8$ (vs. +47.3pp for Qwen, both large and monotonic in N); (iii) BM25 partially recovers performance and also collapses truncation. Surface metrics at $N = 8$ are backbone-agnostic to within noise: tool-call syntax = 1.00, balanced XML = 1.00, non-empty queries = 1.00, action-sequence accuracy = 0.606, BLEU = 1.16, ROUGE-L = 0.21 (Appendix K); the smaller cross-family backbone lowers the absolute ceiling but preserves the diagnostic shape. The transferable lesson is the protocol, not the specific 3B model it was developed on; near-identical surface estimates reflect discrete action-prefix and floor-level BLEU/ROUGE bunching, with overlapping bootstrap CIs (Appendix L).

4 Related Work

RL-trained LLM agents. RL fine-tuning endows LLMs with procedural capabilities SFT does not produce (Zhou et al., 2025, 2024a; Liu et al., 2024; Yao et al., 2023; Zeng et al., 2024). We investigate why these RL-acquired strategies resist distillation.

Knowledge distillation for LLMs. Standard methods (Hsieh et al., 2023; Gu et al., 2024; Jiang et al., 2023) target single-turn transfer; policy distillation and imitation learning (Rusu et al., 2016; Czarnecki et al., 2019; Ross et al., 2011) handle procedural knowledge in low-dimensional action spaces. Our setting differs: the teacher's competence arises from a *multi-turn procedural policy* in the high-dimensional token space of LLMs.

SFT–RL alignment gaps and weight-space analysis. The "alignment tax" (Askell et al., 2021) documents RLHF degrading pre-training capabilities. Jin et al. (2025a) show SFT induces "hard alignment" while RL produces "soft alignment"; Aghajanyan et al. (2021) demonstrate fine-tuning operates in low intrinsic dimension. PiSSA (Meng et al., 2024) initializes LoRA from principal singular components; our SVD alignment ratio measures the overlap of *fine-tuning deltas* with the base model's principal subspace. We extend these by revealing quantitative gaps (2–6$\times$ for RL vs. 1.5$\times$ for SFT) not captured by prior metrics. Extended related work on multi-objective RL and preference learning is in Appendix I.

The evaluation blind spot in agentic distillation. Recent work increasingly distills tool-using capabilities from large models into smaller ones, but evaluation practices vary widely in whether they test actual tool execution. Several works evaluate only format correctness: Jhandi et al. (2025) use ChatGPT-as-judge to score tool-call syntax without executing calls; Yin et al. (2025) evaluate against BFCL format matching, which measures argument-level accuracy but not end-to-end execution; AgentKD (ModelScope Team, 2025) trains on virtual tool-use tasks with pre-defined return values. These evaluations implicitly assume that format accuracy entails execution capability. Our results show this assumption can fail catastrophically: Jaccard = 0.95 (near-perfect format) coexists with em_partial = 0.025 (near-zero task performance). Works that provide real tool access at test time (Kang et al., 2025; Liu et al., 2025) produce more robust evaluations by running students with live retrieval or code tools. To our knowledge, however, these works do not report a single-variable with/without tool-backend ablation that isolates the execution bottleneck on a held-fixed student checkpoint. We propose this diagnostic as a standard component of agentic distillation evaluation.

Comparison to Kang et al. (2025). Kang et al. (NeurIPS 2025) present the closest prior agentic-distillation study: they distill tool-using agents into 0.5–3B students with a first-thought prefix and self-consistent action generation, and evaluate end-to-end with live retrieval and code backends. Our contribution is orthogonal: (i) we inject oracle retrieval into a *held-fixed* SFT checkpoint so that the only variable between the failing and recovering runs is the tool backend, rather than comparing two differently trained students; (ii) we inject teacher-trajectory information segments, which probes whether the student's learned *search strategy* is viable given perfect execution; and (iii) our framing targets the *silent-failure* mode in which format-level indicators and query fidelity remain near ceiling while content-level em_partial collapses—a failure signature we document on both Qwen2.5-3B and Llama-3.2-1B. Combining these interventions reveals that the gap is neither a capacity nor a preference-learning problem but an execution-environment mismatch.

5 Conclusion

Agentic SFT distillation can produce students that pass format-level, tool-call structural checks—perfect tool-call syntax, balanced XML tags, near-perfect query similarity—yet retain less than 6% of the teacher's task performance, and we show this collapse replicates across two backbones and two parameter scales (Qwen2.5-3B and Llama-3.2-1B, spanning two model families and a $\sim$3$\times$ size difference). We have proposed a diagnostic protocol to detect this silent failure mode before pipeline deployment. The protocol proceeds by elimination: characterize the collapse shape, analyze weight geometry for geometric gaps between RL and SFT, and apply causal interventions (oracle and BM25 retrieval) that distinguish strategy acquisition from execution capability. Because the protocol requires only inference-time ablations, it is transferable to any agentic distillation setting where the teacher's competence depends on tool execution, and its signals (truncation scaling, oracle recovery, surface-vs-content metric gap) reproduce across at least two distinct student backbones (§3.6).

Applied to MEM1 (7B RL $\rightarrow$ 3B SFT, with a cross-backbone Llama-1B replication), the protocol yields five findings: (1) multi-objective performance degrades monotonically with no phase transition, setting slope reduction as the distillation target; (2) RL weight changes concentrate 2–6$\times$ supra-random along base principal directions, while SFT recovers only 1.47$\times$—a geometric gap robust across 196 matrices ($p < 1.3 \times 10^{-32}$); (3) DPO fails to close the gap (+1.0pp, power $>$ 0.99 at 5pp), ruling out insufficient training signal; (4) oracle (teacher-retrieval replay) recovers +47.3pp ($p < 0.0001$), upper-bounding what retrieval-side mitigation can achieve and consistent with retrieval execution—not weight alignment or capacity—dominating the gap, with capacity contributing at most $\sim$12–14% and retrieval-quality recovery upper-bounding the remainder; even BM25 retrieval recovers +6–12pp (within the teacher's training document pool), offering an immediately deployable mitigation; (5) the silent-failure signature and behavioral-mismatch diagnosis replicate on Llama-3.2-1B—format-level indicators all at 1.00, action-sequence accuracy 0.606, BLEU 1.16, ROUGE-L 0.21, and oracle retrieval again dropping truncation from 29–86% to 0% while recovering em_partial by $\approx$34pp at $N = 8$—evidence that the finding generalises across model families and parameter scales (§3.6).

5.1 Limitations

Scope and capacity isolation. All experiments use one RL teacher (MEM1, 7B) on one domain (multi-objective RAG QA). Student-side we cover two backbones and two scales (Qwen2.5-3B and Llama-3.2-1B, cross-family, $\sim$3$\times$ size difference) and observe the same silent-failure signature on both (§3.6); direct replication on other teachers and benchmarks remains future work. The 7B full-rank pooled SFT quantifies the capacity contribution at $\approx$12–14% of the teacher–student gap (§3.2), but the full-rank vs. LoRA training-method mismatch makes this a best-effort upper bound.

DPO data scarcity and retrieval oracle. DPO uses only 176 matched pairs (0.4% perfect rate at $N = 8$), a structural consequence of multi-objective collapse, not a remediable sampling issue; power exceeds 0.99 for a 5pp effect (floor 2.7pp). Oracle retrieval uses teacher passages and is thus an upper bound, but BM25 retrieval confirms the finding generalizes beyond the oracle setting (+6–12pp within the teacher's training document pool, truncation $\rightarrow$ 0%).

Ethics Statement

Data. All training data comes from FlashRAG (MIT license) (Jin et al., 2025b) and the MEM1 teacher trajectories (MIT license) (Zhou et al., 2025). The underlying QA datasets (NQ, TriviaQA, HotpotQA, and FlashRAG's multi-hop mixture) contain no personally identifiable information beyond what is already public on Wikipedia.

Computational cost and environmental impact. Our experiments used approximately 50 GPU-hours on an NVIDIA RTX PRO 6000 Blackwell (97 GB, ∼600 W TDP) single-node setup, distributed across Qwen2.5-3B / 7B / Llama-3.2-1B SFT, DPO, and evaluation runs (the 7B full-rank SFT and the Qwen-3B eval sweep dominate the budget). Using the carbon-estimation methodology of Strubell et al. (2019) with an average draw of ∼0.5 kWh per GPU-hour and an estimated grid carbon intensity of ∼0.5 kg CO_2/kWh, this corresponds to roughly 12 kg CO_2-equivalent. No experiments were rerun for cosmetic reasons; rerun counts are bounded by the registry entries summarised in Appendix A and E.

Misuse considerations. Our diagnostic framework exposes that format-level metrics alone can mask complete task failure in agentic distillation. Deployed systems relying on tool-call format validators as safety filters may therefore be vulnerable to silent-failure modes in which the student produces syntactically well-formed tool calls that never reach a valid retrieval backend. We recommend content-level evaluation and end-to-end tool-execution ablations as complementary checks for any distilled agent.

Bias and limitations. Our evaluation is English-only and Wikipedia-centric, inheriting the coverage bias of the FlashRAG corpus. The silent-failure mode we document may manifest differently in multilingual, domain-specific, or non-retrieval agentic settings. All students are trained under a single hash-based train/test split; we did not vary the split seed (see §5.1).

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A Training Details

Table 4 summarizes the hyperparameters for all training stages.

	SFT (pooled)	DPO (per-N)
Base model	Qwen2.5-3B	SFT checkpoint
LoRA rank	32	32
LoRA α	64	64
LoRA target	all linear	all linear
Learning rate	2×10^{-4}	2×10^{-5}
Epochs	3	1
Batch size	4	4
Gradient accum.	4	4
Max seq. length	6144	6144
Optimizer	AdamW	AdamW
Weight decay	0.01	0.01
Warmup ratio	0.03	0.1
DPO β	—	0.05
Training samples	1,118 (partial-credit, pooled $N \in \{2,4,8\}$)	176 raw pairs (matched to $N=4$ bottleneck)
Train/test split	hash-based, test_frac = 0.25	

Table 4: Training hyperparameters for SFT and DPO stages. Both stages use LoRA (rank 32) on Qwen2.5-3B. The SFT stage uses partial-credit trajectories (inclusion: total_em $\geq$ 1); DPO stages use matched-sample preference pairs subsampled to 176 raw pairs.

SFT convergence (3B). The 3B SFT stage converges within 3 epochs with monotonically decreasing training loss (final loss ≈ 0.91). No overfitting is observed on the held-out split.

7B capacity-isolation model. Table 5 summarizes the hyperparameters for the 7B full-rank pooled SFT used in the capacity-isolation experiment (§3.2).

DPO convergence. All three DPO variants ($N \in \{2, 4, 8\}$) train for 1 epoch. Final training losses range from 0.595 (DPO$_{N=2}$) to 0.678

	7B SFT (pooled)
Base model	Qwen2.5-7B
Fine-tuning	Full-rank (no LoRA)
Learning rate	2×10^{-5}
Epochs	2
Batch size	1
Gradient accum.	4 (eff. batch 4)
Max seq. length	6144
Optimizer	AdamW
Precision	bf16
Gradient checkpointing	Yes
Training samples	1,118 (same partial-credit pooled data as 3B SFT)
Training time	19.5 min (560 steps)
Final train loss	0.83

Table 5: Training hyperparameters for the 7B full-rank capacity-isolation SFT. This model uses the same pooled data and inclusion criteria as the 3B LoRA SFT, differing only in model size (7B vs. 3B) and fine-tuning method (full-rank vs. LoRA rank 32). The lower learning rate (2×10^{-5} vs. 2×10^{-4}) and fewer epochs (2 vs. 3) reflect standard practice for full-rank fine-tuning of larger models.

(DPO$_{N=8}$). The rewards/margins are positive but small: 0.207 (DPO$_{N=2}$), 0.100 (DPO$_{N=4}$), 0.056 (DPO$_{N=8}$), consistent with the limited preference signal available from 176 matched pairs. The decreasing margin with N reflects the increasing difficulty of producing contrastive signal at higher objective counts.

B SVD Alignment Per-Layer Distribution

See Figure 3 in the main text for the per-layer SVD alignment boxplot.

C SVD Weight Geometry Details

Definitions. For a base weight matrix $\mathbf{W}_{\text{base}} \in \mathbb{R}^{m\times n}$ with truncated SVD $\mathbf{W}_{\text{base}} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$, retain the top-$k$ singular vectors $\mathbf{U}_k$, $\mathbf{V}_k$. The observed alignment of a fine-tuning delta $\Delta\mathbf{W} = \mathbf{W}_{\text{ft}} - \mathbf{W}_{\text{base}}$ is

$$\alpha_{\text{obs}}(k) = \frac{\|\mathbf{U}_k^\top \Delta\mathbf{W}\,\mathbf{V}_k\|_F}{\|\Delta\mathbf{W}\|_F},$$

and the alignment ratio is $r(k) = \alpha_{\text{obs}}(k)/\alpha_{\text{rand}}(k)$ where $\alpha_{\text{rand}}(k)$ is the mean projection norm over $n = 100$ Gaussian random matrices matched in shape and entry variance to $\Delta\mathbf{W}$. We decompose into U-side ($\|\mathbf{U}_k^\top \Delta\mathbf{W}\|_F/\|\Delta\mathbf{W}\|_F$) and V-side

($\|\Delta\mathbf{W}\,\mathbf{V}_k\|_F / \|\Delta\mathbf{W}\|_F$) components capturing concentration along base output vs. input directions.

Full alignment ratio table. Across all 196 weight matrices of Qwen2.5-7B (28 layers × 7 projection types), Table 6 reports $r(k)$ for the three deltas RL–base, SFT–base, and RL–SFT.

	$k\!=\!64$	$k\!=\!128$	$k\!=\!256$	$k\!=\!512$	$k\!=\!1024$
RL–base	6.48	4.11	<u>2.78</u>	2.01	1.64
SFT–base	2.92	—	<u>1.47</u>	—	—
RL–SFT	3.80	—	1.81	—	—
RL/SFT ratio at matched k:					
$r_{\mathrm{RL}}/r_{\mathrm{SFT}}$	2.22	—	1.89	—	—

Table 6: SVD alignment ratio $r(k)$ for three weight deltas, averaged over 196 matrices. Wilcoxon $p < 1.3{\times}10^{-32}$ at all k. "—" marks k values not computed for SFT/RL–SFT due to compute budget.

Per-matrix dispersion. At $k = 256$ the per-matrix distribution is right-skewed: $r_{\mathrm{RL}} = 2.78 \pm 2.57$, $r_{\mathrm{SFT}} = 1.47 \pm 2.19$ (mean ± std), with a few outlier layers reaching $r > 10$. Only 2–3 layers show $r_{\mathrm{SFT}} > r_{\mathrm{RL}}$ (most notably layer 1 `mlp.down_proj`); in those cases both ratios remain $r > 2$, so the exceptions are high-alignment for both methods rather than SFT-dominant.

Layer-type decomposition. The alignment gap is not uniform. At $k = 256$, MLP layers show stronger RL–base alignment (3.49×) than attention layers (2.25×), while SFT–base alignment is 2.01× for MLP and only 1.07× for attention—SFT attention deltas are nearly indistinguishable from random. The multi-objective capability encoded in attention weights is therefore the most resistant component to SFT transfer.

U-side / V-side asymmetry. For attention layers, U-side alignment dominates V-side (0.31 vs. 0.12 at $k = 256$): RL modifies how attention *writes* its output more than how it reads. For MLP layers the pattern reverses (U-side 0.05, V-side 0.11): RL modifies how MLP layers select input features. A successful distillation method would need to replicate not just the magnitude but the directional structure of RL changes across these layer types.

Same-architecture comparison. Both the RL and SFT models compared here are full-rank fine-tunes of the same Qwen2.5-7B base, so alignment ratios reflect differences in training objective (PPO vs. behavior cloning) rather than dimensional or rank confounds. The distillation experiments in §3 use a separate Qwen2.5-3B / LoRA rank-32 student, reflecting the deployment cost constraint.

D Oracle Retrieval Injection: Implementation and Leakage Audit

Pseudocode. Algorithm 1 describes our oracle retrieval injection. The implementation builds a map from each test prompt to the ordered list of <information> segments in the teacher's best (highest-EM) trajectory for that prompt (from `phase1d_v2_data/raw_trajectories_N*.json`). At inference time we run the student in an interactive loop, halting whenever it emits </search>, and inject the j-th teacher segment in response to the student's j-th search query. The mapping is *positional* rather than query-similarity-based: we do not match on Jaccard or cosine similarity between the student query and the teacher query. This keeps the injection deterministic and reproducible, and it matches the teacher's own ordering of retrieval rounds.

Algorithm 1 Oracle Retrieval Injection (exp_009 / exp_013).

Input: Student prompt p, teacher trajectory T_p (best-EM response), maximum search rounds R
Output: Student completion with oracle-injected retrieval segments
Parse T_p into ordered information segments $[I_1, \ldots, I_k]$ using the regex <information>.*?</information>
$j \leftarrow 0$; gen $\leftarrow$ generate from p with stop strings {</search>, </answer>}
while gen ends with </search> and $j < R$ **do**
  if $j < k$ **then**
    Append I_{j+1} verbatim to gen
  else
    Append a no-hit <information></information> block
  end if
  $j \leftarrow j + 1$
  Continue generation from gen until </search>, </answer>, or `max_new_tokens`
end while
return gen

Concrete parameters: `max_new_tokens` = 4096, `max_search_rounds` = R = 5, greedy decoding (`do_sample =False`), model loaded as a merged LoRA adapter. Both Qwen-3B (exp_009) and Llama-1B (exp_013) use the same injection loop; only the base checkpoint and adapter change.

Leakage Audit

To verify that the oracle retrieval intervention does not trivially leak gold-answer strings, we run a substring audit on a random sample of 50 N=8 evaluation prompts (`scripts/oracle_leakage_audit.py`, seed=42). For each prompt, we concatenate all teacher-injected `<information>` blocks, lowercase and strip punctuation from both the concatenated text and the gold-answer strings (from the FlashRAG evaluation JSON), and record a hit if any gold answer is a substring of the concatenated blocks. As a control, we shuffle teacher passages across prompts (prompt p receives passages from a random other prompt p') and re-run the same check.

Condition	Ratio (gold-answer hit / 50)
Real (teacher's own info blocks)	49 / 50 = 98.00%
Shuffled (derangement) control	17 / 50 = 34.00%
Δ (real $-$ shuffled)	64 pp

Table 7: Oracle retrieval leakage audit on 50 random N=8 test prompts.

Interpretation. The absolute 98% real-hit rate is high, and the audit script's built-in decision threshold ($>30\%$) tags this rate as "non-trivial leakage"; we therefore retain the experiment as an *upper-bound probe*, not as a clean causal estimate. Two independent checks nevertheless bound the leakage contribution: **(i)** the teacher MEM1 policy is trained with PPO on outcome rewards only, and its FlashRAG retriever receives no gold-label supervision at rollout time; **(ii)** under teacher-retrieval replay, student em_full at $N = 8$ remains 0.008, far below the ≈ 1.0 rate expected if passages directly exposed answers. Accordingly, the $+47.3$ pp oracle em_partial recovery should be read as a *strict upper bound* on retrieval-side mitigation, not as a demonstration of perfect answer access. The 64 pp real$-$shuffled gap itself quantifies retrieval topic specificity relative to the derangement baseline.

Limitations. The substring match does not distinguish partial lexical overlap from semantic coverage; 50 samples is a sanity check rather than a full audit. A comprehensive audit covering all splits and semantic paraphrase matching is deferred to camera-ready.

E Cross-Backbone: Llama-3.2-1B SFT Hyperparameters

Table 8: Llama-3.2-1B LoRA SFT hyperparameters (exp_013, cross-backbone W2).

Hyperparameter	Value
Base model	`meta-llama/Llama-3.2-1B`
LoRA target	q,k,v,o,gate,up,down_proj
LoRA rank	32
LoRA α	64
LoRA dropout	0.05
Optimiser	AdamW
Learning rate	2×10^{-4}
LR scheduler	cosine, warmup 10%
Weight decay	0.01
Batch size	1
Gradient accum.	8 (eff. batch 8)
Epochs	3
Max seq. length	6144
Precision	bf16
Gradient checkpointing	Yes
Seed	42
Chat template	Llama-3.1 (manual)
Pad token	`eos_token`
Training samples	1,118 (same pooled split as 3B)
GPU	NVIDIA RTX PRO 6000 Blackwell (97 GB)
Training wall-clock	10.1 min (603 s)

This configuration mirrors the Qwen2.5-3B SFT in Table 4 on every axis that could confound the cross-backbone comparison (LoRA rank/α, epoch count, learning rate, data, split seed). Only the base model, chat template, and effective batch differ—the effective batch is 8 (batch 1 $\times$ grad_accum 8) vs. 16 for Qwen-3B—because Llama-3.2-1B is roughly $1/3$ the parameter count of Qwen2.5-3B and gradient accumulation was re-tuned for throughput on the single-GPU Blackwell box. We verified that the Llama student still converges within 3 epochs (final train loss $\approx$ 0.74) on the same partial-credit pooled data.

F Responsible NLP Research Checklist

We summarise the ACL Responsible NLP Research Checklist for the present submission.

A. Scientific artifacts. We use FlashRAG (Jin et al., 2025b) (MIT license), MEM1 teacher checkpoints and trajectories (Zhou et al., 2025) (MIT license), Qwen2.5-3B / 7B (Apache 2.0), and Llama-3.2-1B (Meta Llama 3 Community License, which imposes use-case and commercial-use restrictions). We use all artifacts consistent with their intended research use. We release code and evaluation scripts under the same MIT license as FlashRAG. All derived artifacts (distilled adapters, eval outputs) inherit the most restrictive upstream license.

B. Computational experiments. See Ethics Statement (*Computational cost and environmental impact*) and Appendices A, E. We report per-experiment GPU time, effective batch size, and total wall-clock for all training runs. Evaluation runs use max_new_tokens = 4096 throughout (with the truncation confound of max_new_tokens = 2048 documented in Appendix J). All reported hyperparameters come from a single run per configuration; we report exact seeds (split seed 42, training seed 42) and do not perform hyperparameter search beyond the grid reported in Table 4.

C. Human annotators. Not applicable. No human annotation, crowdsourcing, or user studies were conducted.

D. AI assistants. We used large-language-model writing assistants (ChatGPT, Claude) for language polishing of the manuscript. All experimental design, code, analysis, figures, and claims are the authors' own work; no text was inserted by an AI assistant without author verification and rewriting.

G Activation Probing Details

We extract activation differences $\Delta\mathbf{h}_l = \mathbf{h}_l^{\mathrm{RL}} - \mathbf{h}_l^{\mathrm{base}}$ at layers $l \in \{0, 3, 7, 11, 15, 19\}$ for $n = 300$ prompts, projecting onto the top-k principal directions ($k \in \{64, 256\}$) of the base model's weight matrices. Eight feature variants are tested: raw activations $\mathbf{h}^{\mathrm{RL}}$, $\mathbf{h}^{\mathrm{base}}$, their difference $\Delta\mathbf{h}$, and top-k projections thereof.

Linear probes. Binary classification (per-objective EM above/below median) achieves accuracy 0.72, below the majority-class baseline 0.76. Regression probes yield negative R^2 across all 55 (feature, layer, k) combinations.

MLP probes. Two-layer MLPs (hidden dimension 128, ReLU) trained on the same features predict per-objective EM as a continuous target (Spearman ρ). MLP probes lose to linear probes in 47/55 cases (85.5%; paired $t = -8.19$, $p = 4.83 \times 10^{-11}$, Cohen's $d = -1.105$). Mean ρ: -0.011 (MLP) vs. 0.073 (linear).

Random-feature baseline. We construct 5 random orthonormal bases × 6 layers × 2 k values. The maximum $|\rho|$ under random features is 0.224; 52/55 (94.5%) of real linear probe $|\rho|$ values fall within this noise band. The per-objective signal in $\Delta\mathbf{h}$ is indistinguishable from random projections.

H DPO Ablation Details

We train three per-N DPO variants on Qwen2.5-3B (LoRA rank 32), each initialized from the pooled SFT checkpoint and trained on N-specific preference pairs. The matched-sample constraint (176 raw pairs, subsampled to the $N = 4$ bottleneck) is critical for fair cross-N comparison but severely limits data volume: at $N = 8$ only 0.4% of teacher trajectories achieve perfect scores. This scarcity is a structural consequence of multi-objective collapse.

Model	$N=2$	$N=4$	$N=8$
SFT shared (3B)	0.112	0.084	0.025
SFT full-rank (7B)	0.171	0.147	0.079
DPO $N=2$	0.128	0.095	0.031
DPO $N=4$	0.128	0.099	0.026
DPO $N=8$	0.130	0.082	0.035
RL teacher (7B)	0.59	0.54	0.455

Table 9: Full em_partial evaluation matrix (max_new_tokens = 4096). Best $N = 8$ DPO: +1.0pp ($p = 0.163$).

The best result—DPO$_{N=8}$ at eval $N = 8$—yields $\Delta = +1.0$pp ($p = 0.163$, 95% CI $[-0.9, +2.9]$pp), far below the pre-registered practical-significance threshold of +5pp. A post-hoc power analysis confirms an informative null: with $n = 125$ paired test prompts and observed $\sigma_{\mathrm{diff}} \approx 10.8$pp, we achieve > 0.99 power at 5pp and 0.80 power at 2.7pp. No systematic N-matching effect is observed: DPO$_{N=2}$ performs comparably to DPO$_{N=8}$ across conditions. Both students retain <8% of teacher performance at $N = 8$ (SFT 0.025, DPO$_{N=8}$ 0.035), and neither achieves em_full > 0 at $N \geq 4$.

I Extended Related Work

Multi-objective RL for LLMs. Recent work explores multi-objective optimization in LLM alignment: Rewarded Soups (Rame et al., 2024) interpolates weights fine-tuned on diverse rewards, MODPO (Zhou et al., 2024b) extends DPO to multiple preference dimensions, and fine-grained human feedback (Wu et al., 2024) decomposes rewards into per-aspect signals. These methods address *training* with multiple objectives; our work addresses the complementary problem of *distilling* multi-objective RL policies.

Preference learning and DPO. DPO (Rafailov et al., 2023) and extensions (IPO (Gheshlaghi Azar et al., 2024), KTO (Ethayarajh et al., 2024), RAFT++ (Dong et al., 2023)) provide offline alternatives to RLHF. We use DPO as a stronger-than-SFT baseline; its failure (+1.0pp, not significant) suggests behavioral mismatch rather than insufficient signal richness. Following Karl et al. (2024), we frame this as a diagnostic contribution.

J Truncation Confound Analysis

An important methodological episode shaped our analysis. An earlier evaluation used max_new_tokens = 2048, under which the DPO advantage appeared significant (Table 10). Re-evaluating SFT at 4096 tokens revealed the original significance was largely an artifact.

	em_partial	trunc.	Δ vs SFT	p
SFT at max_new_tokens = 2048 (confounded):				
SFT shared	0.011	93.6%	—	—
DPO $N=8$	0.035	—†	+2.4pp	< 0.001
SFT at max_new_tokens = 4096 (corrected):				
SFT shared	0.025	83.2%	—	—
DPO $N=8$	0.035	∼84%	+1.0pp	0.163

Table 10: Truncation confound at $N=8$. Extending SFT's budget to 4096 tokens lifted em_partial from 0.011 to 0.025 and revealed the DPO advantage is not significant. †DPO was always evaluated at 4096.

At 2048 tokens, 93.6% of SFT outputs were truncated before the <answer> tag; extending to 4096 allowed ∼10% more to complete. DPO's score was unchanged—its apparent advantage came from SFT being disproportionately penalized. A pilot attempt to suppress search tokens via bigram blocking was inconclusive: the model generated tag variants (malformed closing tags, inserted whitespace) that bypassed token-level suppression, suggesting the behavioral pattern is encoded above individual token sequences. This confound is specific to RL-distilled models whose procedural strategies produce verbose outputs; we report it as a methodological caution for future work.

K Full Silent Failure Metric Breakdown

	Format-level	Content-level
Qwen-2.5-3B	≈ 1.00 (all checks)	em_p=0.025
Llama-3.2-1B	≈ 1.00 (all checks)	em_p=0.014

Table 11: Silent failure signature replicates across backbones. Format-level indicators (tool-call syntax, balanced XML tags, non-empty queries) ≈100% for both; content-level em_partial near-floor at $N = 8$ (Qwen query Jaccard = 0.95, also near-ceiling). Dispersion analysis in Appendix L.

L Surface Metric Dispersion Analysis

Near-identical cross-backbone point estimates on action-sequence accuracy, BLEU, and ROUGE-L (Table 11) raise a concern that the agreement is coincidental. To rule this out we compute per-sample 95% bootstrap confidence intervals (10,000 resamples, seed 42) on the two surface metrics with per-sample access—action-sequence accuracy and ROUGE-L—and on the task-level em_partial across $N \in \{2, 4, 8\}$ for every backbone × retrieval configuration. Results are in Table 12.

The overlap has a mechanistic explanation: (i) action-sequence accuracy is dominated by a shared initial "search" action prefix that both students emit deterministically, producing discrete mass at identical values; (ii) BLEU (corpus-level, not bootstrapped here) sits near the floor for both students on sacrebleu's 0–100 scale (1.14 Qwen, 1.16 Llama), small relative to the teacher's 0.455 em_partial signal; (iii) ROUGE-L at ≈ 0.21 reflects shared tool-call markup tokens rather than substantive content overlap. The consistency is therefore a property of the silent-failure regime, and does not affect any inference in the main text (all causal claims rest on the oracle-recovery contrast in §3, not on surface-metric equality). The same bootstrap sweep also quantifies em_partial dispersion for every cell of the cross-backbone table: Llama-1B oracle retrieval lifts em_partial to 0.351 $[0.318, 0.384]$ at $N = 8$ (vs. 0.014 $[0.005, 0.025]$ no-API), non-overlapping CIs that anchor the +33.7pp oracle-recovery claim.

Backbone	Retrieval	Metric	Split	N	Mean	95% CI
Qwen-3B	no-API	action_seq_acc	N=8	125	0.606	[0.595, 0.616]
Qwen-3B	no-API	ROUGE-L	N=8	125	0.214	[0.203, 0.225]
Llama-1B	no-API	action_seq_acc	N=8	125	0.606	[0.595, 0.616]
Llama-1B	no-API	ROUGE-L	N=8	125	0.212	[0.201, 0.223]
Qwen-3B	no-API	em_partial	N2	251	0.112	[0.084, 0.139]
Qwen-3B	no-API	em_partial	N4	131	0.084	[0.059, 0.111]
Qwen-3B	no-API	em_partial	N8	125	0.025	[0.013, 0.038]
Llama-1B	no-API	em_partial	N2	251	0.074	[0.052, 0.098]
Llama-1B	no-API	em_partial	N4	131	0.063	[0.042, 0.086]
Llama-1B	no-API	em_partial	N8	125	0.014	[0.005, 0.024]
Llama-1B	oracle	em_partial	N2	251	0.410	[0.367, 0.454]
Llama-1B	oracle	em_partial	N4	131	0.401	[0.357, 0.445]
Llama-1B	oracle	em_partial	N8	125	0.351	[0.320, 0.382]
Llama-1B	BM25	em_partial	N2	251	0.135	[0.106, 0.167]
Llama-1B	BM25	em_partial	N4	131	0.094	[0.069, 0.120]
Llama-1B	BM25	em_partial	N8	125	0.066	[0.050, 0.083]

Table 12: Bootstrap 95% CIs (10,000 resamples, seed 42) for per-sample surface metrics and task em_partial across backbones and retrieval modes. Qwen-3B per-sample action-sequence and ROUGE-L values come from the stored W1 surface-metrics log; Llama-1B values are recomputed via the same extraction from `exp_013` eval details. At $N = 8$ the surface-metric CIs for the two backbones overlap almost entirely (action_seq_acc 0.606 $[0.595, 0.616]$ for both; ROUGE-L 0.214 $[0.204, 0.224]$ vs. 0.212 $[0.202, 0.224]$), confirming that the near-identical means reported in §1 and §3 reflect a genuinely near-identical distribution rather than a point-estimate coincidence.